Ordinal regression based on ensemble of classifiers

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Introduction

In many classification problems in research and in industry, labels possess a natural order. Standard multiclass classification methods treat these ordinal labels as nominal, ignoring the inherent order relation in the data and therefore misclassifications are considered as binary penalties. To address this limitation, we propose a method for multiclass classification that explicitly accounts for the ordinal nature of such problems. Our model aims to increase penalty for larger deviations from the ground truth.

Problem setting

Consider a multi-classification task with K ordered classes and a set of T classifiers. Let $X = [x^1, \dots, x^n] \subset \mathbb{R}^{p\times n}$ be the labeled data and $\bar{\imath} \in [1, K] := \{1, \cdots, K\}$ the classification of the point x^i.

Each classifier $t \in [1, T]$ assigns each point $x^i \in X$ in a class. Based on this, for each point x^i and class k we define the vector $y_k(x^i) \in \{0, 1\}^T$. The t-th entry of $y_k(x^i)$ is equal to 1 if the classifier t assigns x^i as k and 0 otherwise.

Literature model

The model proposed by [1] aims to find a linear combination of the classifiers which will give an order of the labels, that is, to find $\omega^* \in \mathbb{R}^T$ and $\xi^* \in \mathbb{R}^n$ that solves the optimization problem

$$\begin{aligned} \min_{\omega,\xi} \quad & \textstyle\sum_{i=1}^n \xi_i \qquad \text{(L)} \\ \text{s.t.} \quad & \omega^\top y_{\bar{\imath}}(x^i) \geq \omega^\top y_j(x^i) - \xi_i + \epsilon, \quad i \in [1, n],\ j \in [1, K]\backslash\{\bar{\imath}\}, \\ & \omega_t \in [0, 1], \quad t \in [1, T], \quad \xi_i \geq 0, \quad i \in [1, n]. \end{aligned}$$

where $\epsilon > 0$. Given a new point x with an unknown class, its classification is determined by

$$\arg\max_{k\in[1,K]} \omega^\top y_k(x).$$

Our model

The idea in this work is to impose a greater misclassification penalty, as the predicted classification deviates further from the true classification. The following model yields that:

$$\begin{aligned} \min_{\omega,\xi} \quad & \textstyle\sum_{i=1}^n \xi_i \qquad \text{(O)} \\ \text{s.t.} \quad & \omega^\top y_{\bar{\imath}}(x^i) \geq \omega^\top y_j(x^i) - \frac{\xi_i}{|j - \bar{\imath}|} + \epsilon, \quad i \in [1, n],\ j \in [1, K]\backslash\{\bar{\imath}\}, \\ & \omega_t \in [0, 1], \quad t \in [1, T], \quad \xi_i \geq 0, \quad i \in [1, n]. \end{aligned}$$

Experiments

The experiments were run on `julia` using `JuMP` modeling package and `Ipopt` solver and using the datasets from [2]. Use used $T = 7$ `DecicionTreeClassifiers`, each one receiving 1% of the data to train on. The mean absolute error (MAE) was calculated for each individual classifier, and resumed on the Classifiers column on Table 1 taking the mean and standard deviation over the seven classifiers. The literature model and our model were trained on the same 7% of the data used to train the individual models using $\epsilon = 0.1$, and their MAE were calculated on the remaining 93% of the data.

Our model has better results on 8 and equal results on 7 out of 17 datasets.

Table 1: Summary results. Values for MAE listed as mean±sd.

dataset	K	Classifiers	(L) Model	(O) Model
automobile	6	.4590±.0875	.1748±.5503	**.1503**±.4777
balance-scale	3	.4984±.0436	.2404±.5879	**.2370**±.5834
bondrate	5	.6210±.1202	**.3055**±.5650	.3407±.6057
car	4	.2399±.0103	**.2188**±.4960	**.2188**±.4960
contact-lenses	3	.4488±.2834	**.2146**±.5822	**.2146**±.5822
ERA	9	1.4756±.0678	1.3552±1.3528	**1.3540**±1.356
ESL	9	.3709±.0433	.2586±.4918	**.2568**±.4741
eucalyptus	5	.4635±.0272	.3139±.5276	**.3136**±.5249
LEV	5	.5024±.0392	.4331±.6265	**.4070**±.5603
newthyroid	3	.0830±.0144	**.0097**±.0978	**.0097**±.0978
pasture	3	.3974±.1202	**.1466**±.3538	**.1466**±.3538
squash-stored	3	.3787±.0969	**.1416**±.4038	**.1416**±.4038
squash-unstored	3	.2331±.0674	**.0608**±.2390	**.0608**±.2390
SWD	4	.4938±.0323	**.4337**±.5509	.4346±.5492
tae	3	.7693±.1125	.6168±.7439	**.6104**±.7449
toy	5	.2444±.0608	**.0749**±.2633	**.0749**±.2633
winequality-red	6	.4684±.0299	.3766±.5599	**.3684**±.5556

Future work

- Introduce nonuniform misclassification penalty;
- Introduce *per class* misclassification penalty;
- Work on the limitation of the $\omega^\top y_j(x^i)$ value;

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References

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